Ashley Wu

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EDUCATION

Stanford University *Sep.2023 - Present*
- Ph.D. Candidate in Bioengineering, Minor in Computer Science (Expected Jun. 2029)
- M.S. in Electrical Engineering; GPA: 4.13/4.00

Tsinghua University *Aug.2019 - Jun.2023*
- B.Eng. of Biomedical Engineering; GPA: 3.96/4.00, ranking 1/15

Selected Coursework
- **3D Vision, Machine Learning & AR/VR:** Deep Learning for Computer Vision (A), Robotic Perception (A+), Machine Learning (A), Virtual Reality (A), Mining Massive Data Sets (A+)

RESEARCH INTERESTS

3D Computer Vision, 3D Scene Reconstruction, Gaussian Splatting, Novel View Synthesis, AR/VR

TECHNICAL SKILLS

Programming Python, C/C++, C#, MATLAB, JavaScript
Deep Learning & Computer Vision PyTorch, OpenCV, Open3D, scikit-learn, NumPy/SciPy
3D Graphics & Extended Reality Unity, OpenXR, Mixed Reality Toolkit (MRTK), GLSL
Development Tools Linux, Git/GitHub, Docker, ROS 2, Jupyter, Apache Spark

SELECTED PUBLICATIONS

1. **Y. Wu**, *et al.*, "OffTrackGS: Geometry-Constrained Regularization and Ground-Truth-Free Evaluation for Free-Viewpoint Endoscopic Reconstruction," *IEEE Transactions on Visualization and Computer Graphics (IEEE-TVCG)*, Under review.
2. T. El Chemaly, ..., **Y. Wu**, *et al.*, "Seeing Through the Surface: Evaluating Visualization Techniques for Depth Perception in Augmented Microscopy," *IEEE-TVCG*, Accepted.
3. **Y. Wu** and B. Daniel, "Optimizing Projector Pose Estimation for Selective Lighting in Optical See-Through Mixed Reality Systems," *2024 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, IEEE, 2024.

RESEARCH & DEVELOPMENT EXPERIENCES

Free-Viewpoint 4D Gaussian Splatting for Dynamic Surgical Scene Reconstruction **Stanford University**
Mentor: Prof. Bruce L. Daniel *Feb. 2026 – Present*
- Developed OffTrackGS, a 4D Gaussian Splatting framework for reconstructing dynamic surgical scenes from endoscopic video, targeting novel-view rendering beyond the recorded camera trajectory.
- Designed Geometry-Constrained Sideview Regularization to synthesize off-trajectory training views and regularize their geometry, with an anisotropy penalty to prevent degenerate solutions.
- Introduced Round-Trip View Consistency, a ground-truth-free evaluation protocol for quantitative assessment of reconstruction fidelity at unobserved viewpoints.
- Achieved more complete and visually coherent side-view reconstructions, with statistically significant improvements in Round-Trip View Consistency across a range of viewing-angle offsets.

State Estimation-Powered Augmented Reality Guidance for Breast-Conserving Surgery **Stanford University**
Mentors: Prof. Bruce L. Daniel, Prof. Brian Hargreaves, Prof. Jacqueline Tsai *Jan. 2025 – Present*
- Developed a real-time AR guidance system that uses radar-based distance measurements to register preoperative 3D models with deformable patient anatomy.
- Designed a particle filter-based state estimation method to improve spatial registration under soft-tissue deformation and measurement noise, reducing deformation-induced misalignment from 15 mm to 3 mm in experimental evaluation.
- Collaborated with clinical experts to evaluate the prototype, demonstrating localization accuracy comparable to human expert performance.

Joint Hand Segmentation and Depth Estimation from Monocular Video **Stanford University**
Deep Learning for Computer Vision Final Project *Apr. 2024 – Jun. 2024*
- Proposed a unified architecture with a shared encoder and task-specific deconvolutional decoders to jointly perform hand segmentation and depth estimation from monocular RGB video.
- Trained the model with VGG-based initialization using combined segmentation (F1) and depth (MSE) losses, and performed hyperparameter tuning for stable optimization.
- Compared the method against a Fully Convolutional Network baseline, achieving improved depth estimation performance (accuracy: 0.82 vs. 0.67) while maintaining competitive segmentation accuracy (F1 score: 0.91).

AWARDS & HONORS

Stanford Interdisciplinary Graduate Fellowship Stanford University *2026*
Stanford Graduate Fellowship Stanford University *2023*